SOFTWARE TEST PLAN (STP)

Execution Results Addendum — Snake Game

Supplement to the existing 16-page Software Test Plan

Project	Snake Game — Software Engineering Mini-Project
Team	Charan, Vrunda, Ved, Tushar
Update	Test execution status and environment record
Automated test result	68 passed / 0 failed
Evidence	reports/Automated_Test_Results_68_Tests.png
Status note	This addendum supplements the original STP; it does not replace approvals or unverified security checks.

Document control

Prepared as an execution-status supplement. The original STP is revision 1.0 dated 12-10-2026 and shows planned test cases as Not Run. This addendum records results reported during project verification and identifies checks that still need evidence.

1. Execution summary

The complete CTest run executed 68 discovered tests. The captured terminal screenshot reports: “100% tests passed, 0 tests failed out of 68” and total real test time of 25.14 seconds. The run was performed in the Windows PowerShell environment after adding the SFML 2.6.2 runtime folder to the process PATH.

Automated test result

Metric	Result
Total discovered tests	68
Passed	68
Failed	0
Pass rate	100%
Elapsed time	25.14 seconds

2. Test execution status

The following manual outcomes were reported by the project team in addition to the automated CTest run. Where the original test procedure requires evidence that has not been confirmed, the limitation is explicitly retained.

Test case	Reported result	Execution note / qualification
TC-01	Pass	Startup and initial display checks reported complete.
TC-02	Pass	Movement, valid turns and reversal rejection reported complete; automated movement tests passed.
TC-03	Pass	Food placement, detection and growth reported complete; automated food tests passed.
TC-04	Pass	Score update and replacement food reported complete; automated score tests passed.
TC-05	Pass	Boundary/self-collision and departing-tail behaviour reported complete; automated collision/integration tests passed.
TC-06	Pass	Restart/reset checks reported complete; automated restart tests passed.
TC-07	Pass — manual	Safe close/shutdown manually reported successful. Sanitizer-enabled execution required by original test steps has not been confirmed.
TC-08	Pass	Three 60-second runs reported at approximately 60 FPS each; exceeds the ≥30 FPS criterion.
TC-09	Pass	External reviewer reportedly identified controls and used the game without assistance.
TC-10	Pass	Ten consecutive start/game-over/restart cycles reportedly completed without crash or freeze.
TC-11	Pass	Module responsibilities reviewed and reported identifiable in source.
TC-12	Pass	Teammate reportedly completed a fresh clone, build and launch successfully.
TC-13	Pass — manual	Unsupported-key, invalid-state and direction-reversal checks reported successful. Sanitizer-enabled run in the original procedure has not been confirmed.
TC-14	Partial	Static source searches reportedly found no command-execution API matches and no unrelated-file-access API matches. Full Process Monitor trace through a complete round/restart/close has not been confirmed.

Important: TC-07 and TC-13 are reported as manual passes only; their sanitizer sub-steps remain unverified. TC-14 remains Partial and SR-02/SR-03 must not be marked fully passed until the complete review and runtime file-activity trace are recorded.

3. Test environment

Item	Recorded environment
Operating system	Windows 11 Home Single Language, build 26300 (64-bit)
Computer	HP Victus 15-fa1xxx
CPU	13th Gen Intel Core i5-13420H
Memory	16 GB RAM (approximately 16,025 MB reported)
GPU	NVIDIA GeForce RTX 4050 Laptop GPU and Intel UHD Graphics
Compiler	Microsoft C/C++ compiler 19.52.36815 for x64 (Preview)
Build system	CMake 4.3.1 (Visual Studio bundled CMake)
Language standard	C++17
Graphics library	SFML 2.6.2
Test framework	GoogleTest plus standalone snake movement test target
Test command	CTest: --test-dir build -C Debug --output-on-failure
Runtime setup	PowerShell PATH included C:\Libraries\SFML-2.6.2\bin before CTest execution.

4. Evidence and remaining actions

Evidence saved: reports/Automated_Test_Results_68_Tests.png — screenshot of the final CTest output showing 68/68 passed and 0 failed.

Before final submission:

1. Record the test execution date, tester names and any additional manual screenshots/logs in the team's execution log.

2. Run the sanitizer-enabled checks required by TC-07 and TC-13, or explicitly document that those sub-steps were not performed.

3. Complete TC-14 by reviewing all file-open/read/write calls and capturing a Windows Process Monitor trace through a full gameplay round, restart and close. Attach the trace or a concise reviewed evidence record.

4. Update both the SRS Section 8 RTM and the STP Section 10 RTM so their statuses and comments agree with the verified evidence. Keep SR-02 and SR-03 incomplete until TC-14 is complete.

5. Confirm instructor/team approvals and update document revision history; approvals are not inferred by this addendum.

5. Conclusion

The automated test suite completed successfully with 68 of 68 tests passing. Reported manual verification also indicates successful functional, performance, usability, reliability, maintainability and fresh-clone checks. Security verification is not fully closed: sanitizer execution for TC-07/TC-13 and the full runtime file-activity trace for TC-14 still require confirmation. Therefore, the automated suite result is 100% pass, but the overall test plan should not be declared fully closed until those remaining checks and document approvals are addressed.

End of execution update